CISC 4020 L1 Bioinformatics

Instructor: Jose Barba, jbarbamontoya@fordham.edu

Office Hours: Mondays and Thursdays 17:15–18:15, in person at FCLC or virtually, by appointment.
Course Teaching Assistant: TBD, email@fordham.edu.
Course Repository: https://github.com/josebarbamontoya/fordham_bioinformatics.
Email Times: I check email regularly on weekdays and several times on weekends and generally respond within 24–48 hours.
Class Meeting Time and Location: Mondays and Thursdays 16:00–17:15, LL 602.

Course Credits
4 credits.

Course Description
This class is designed to introduce students to the basic concepts and tools of bioinformatics using genomics as a guiding principle. At its core, bioinformatics is all about using computational tools to solve biological problems. Genomics represents a major area of bioinformatics; therefore, the data and tools of this field are emphasized. However, students will be introduced to other areas and be given a chance to conduct independent research in their areas of specific interest.

The course begins with the fundamentals of UNIX/Linux operating systems and working on remote servers. Students then learn to use programming languages and tools, including Shell, Python, and R, to explore and manipulate data from the command line. The course advances to the use of algorithms, databases, and computational methods for managing and analyzing data, allowing the students to design their own data analysis pipeline. Class time focuses on hands-on demonstration and lab exercises designed for the students to fully incorporate the skills they are learning.

New material will generally be introduced before class through readings and guided tutorials. This allows class time to be used for active learning, in which students engage directly with the material and collaborate with their peers and the instructor while developing the conceptual, computational, and interdisciplinary communication skills needed in bioinformatics.

Learning Objectives
- Build core computational skills
- Introduce students to classic and innovative concepts and tools that are the basis of bioinformatics
- Build solid interdisciplinary knowledge of biology and computer science

Materials for the Course
- A computer capable of working in UNIX/Linux operating systems
- Access to journals for research based on primary literature

Additional Readings
This is a four-credit class, so you are expected to make up an extra-hour' worth of class time with work outside of class. Since everyone comes with difference deficits, I expect students will use this extra time to read up on parts of the class that they need more background in. Here are some sources to get you started:
- Terence A. Brown (2002) *Genomes*, 2nd edition. Wiley-Liss (An older book but an excellent resource for the basics of molecular genetics and genomics. It is available for free from NCBI Bookshelf: https://www.ncbi.nlm.nih.gov/books/NBK21128/.)
- Richard Durbin, et al. (2005) *Biological Sequence Analysis: Probabilistic Models of Proteins and Nucleic Acids*. Cambridge University Press. (A classic, and an excellent discussion of the

statistical underpinnings of the algorithms used in bioinformatics. Available for free as a PDF from the library.)
- Ziheng Yang (2014). *Molecular Evolution: A statistical Approach*. Oxford University Press. (An advanced textbook for students and professional researchers, in the fields of bioinformatics and computational biology, phylogenetics, population genetics, and statistical genomics.)
- Jonathan Pevsner (2015) *Bioinformatics and Functional Genomics*, 3rd Edition. Wiley-Blackwell. As to the genomics and theory, you will find most of the material from the class also covered in this book. (Available in the Library at both Rose Hill and Lincoln Center.)

Grade / Evaluation Basis
- Class participation: 10%
- Exams: 20%
- Lab performance: 30%
- *-Omics* paper presentation: 10%
- Bioinformatics research paper presentation: 10%
- Project repository: 10%
- Final Exam: 10%

Attendance
There is no attendance requirement for the class; however, all graded items have an in-class component. For this reason, if a student must miss three or more classes, withdrawal will likely be recommended. Because this is an X-block course, a week of material is covered in each class, and missing three weeks is roughly one third of the class. Students are responsible for all material covered, and lab group members should help keep students apprised of information.

Class Participation
At its minimum, participation means remaining awake and alert during the entire class without succumbing to technical distractions or daydreaming. At its best, it means asking clarifying questions that benefit both the individual and the class, and engaging with lab groups through collaborative problem-solving.

Exams
All exams are open-notes, and students may bring hard copies of any notes they have from the class including slides, Cheat Sheets, homework assignments, and labs. It is regrettable that I must insist on only using hard copies and no other devices or online sources, it is too tempting for students to cheat when using online resources. Students may not use any device during an exam except the computer terminal. The student is only allowed to use the web browser to fill in the exam form, otherwise the students are expected to only use the Linux terminal to complete the exams.

Lab Performance and Groups
Most classes involve practical individual and group work. Grades will split evenly between individual and group contributions. Labs are graded on completion. Groups of three and five will be assigned during the first class, but students will have an opportunity to switch groups any time before the second class ends. After this, the general policy will be that students may not switch groups. Group lab grades are considered both on an individual contribution and group performance level. There are no make-up labs. Missing a class results in a zero for the lab, but these may be dropped if no more than two classes are missed and considered within the context of group performance.

-*Omics* Research Paper Presentation
Each student will choose an *-omics* paper that they find innovative, exciting, relevant to their academic interests. Students will give a 3-minute lightning presentation providing a concise overview of the research question addressed, the key findings, the computational methods employed, and the current limitations of the approach in terms of its application or accessibility. Two weeks before the presentations, I will provide detailed guidance on the expectations and requirements.
<u>Potential topics for *-omics* research paper presentations</u>

1. NGS platforms
2. Long read sequencing
3. Single molecule sequencing
4. Single cell sequencing
5. Direct sequencing of RNA
6. Contact mapping
7. Epigenomics
8. Pan-genomics
9. Metagenomics
10. Metabolomics
11. Spatial transcriptomics
12. Gene/genome editing
13. Genomics and artificial intelligence

Bioinformatics Applications Paper Presentation
This presentation will be completed in groups of five and should be 10-minute long. Each group will select a journal article from the 'journal_articles' folder in the course repository as the basis for their presentation. These articles represent a variety of Bioinformatics applications that students may find of personal/academic interest. In addition to the presentation, students must write an outline for the presentation to be submitted one week before the presentation date. The outline counts for 20% of the presentation grade. After the first exam, I will go over in detail what is expected for the presentations.

Project Repository
The main goal of these projects is to provide students with hands-on experience developing well-formulated research questions that can be addressed using -omics data, selecting appropriate computational methods, and creating fully documented and reproducible computational pipelines. Students will work in groups of three. Each group will receive a dataset and instructions outlining the required analyses one month before the project deadline. Groups will formulate a research question, select and implement appropriate computational methods, develop the necessary analysis pipeline, and create a detailed tutorial documenting the computational steps and workflow.

The project repository will consist of a detailed tutorial, which will be shared on the student's GitHub account. Throughout the course, students will work incrementally to complete the project, adding and testing new steps to the pipeline as they are introduced in class. Each work group will deliver a 10-minute presentation on their project. The presentation will account for 30% of the project repository grade. The submitted GitHub tutorial/repository must meet the following requirements:

1) An "about" section explaining the research goal, why the specific approach presented was chosen, and any alternatives that should be considered. If alternative approaches were tested and rejected, this should be described.
2) For every step of the analysis, provide a description of the tool being used, the available options, any caveats to using the tool, and the exact command used in the project. Even trivial steps like text/data manipulations should be documented.
3) Where relevant, elaborate on any challenges encountered, whether trivial or significant.
4) A brief bibliography of citations for the computational tools used.

5) There is no length requirement for the resulting document, but it must be detailed enough for anyone to fully reproduce your research using the tutorial. It must clearly describe all necessary steps.

Extra Credit
The syllabus provides a complete and final description of all ways to earn credit. Additional or alternative extra credit projects will not be considered or assigned.

Special Cases
Students facing recurring difficulties (e.g., health, family issues, quarantine) should notify the instructor promptly for accommodations. Grades and deadlines cannot be adjusted retroactively.

Academic Integrity
Violations include plagiarism, cheating, or facilitating cheating. Collaborative work is encouraged, but all members must contribute equally and credit must be given. The Fordham University policy is detailed here: https://www.fordham.edu/info/25380/undergraduate_academic_integrity_policy. First violations typically result in an F on the assignment; repeated offenses may result in course failure or expulsion.

AI Policy
Limited usage of generative AI tools may be allowed for specific assignments in this course, enabling exploration of ideas, complex data analysis, and creative solution development, when explicitly permitted by the instructor. When using these tools, it is mandatory to clearly indicate the sections of your work that were generated using them for proper attribution and transparency and indicate the prompts and software versions that were used. It is critical to adhere to ethical standards by refraining from activities like plagiarism or creating misleading content. Additional guidelines or restrictions will be provided for specific assignments.

Accommodations
If you are a student with a documented disability and require academic accommodations, please register with the Office of Disability Services for Students (ODS) in order to request academic accommodations for your courses. Please contact the main ODS number at 718-817-0655 to arrange services. Accommodations are not retroactive, so you need to register with ODS prior to receiving your accommodations.

Grade Scale
93.0−100.0 A
90.0−92.9 A–
87.0−89.9 B+
83.0−86.9 B
80.0−82.9 B–
77.0−79.9 C+
73.0−76.9 C
70.0−72.9 C–
65.0−69.9 D
≤ 65.0 F

Course Schedule (Tentative)

Date	Class #	Topics	Notes
Aug 27	1	– Course introduction and overview – Student backgrounds and interests	
Aug 31	2	– Introduction to bioinformatics	
Sep 3	3	– Introduction to genomics	
Sep 7*	4	– Reproducibility and organization in computational biology – Introduction to GitHub	*Class meets on Tuesday, Sep 8 (Monday class schedule)
Sep 10	5	– Introduction to UNIX – Basic navigation in the terminal	
Sep 14	6	– Data manipulation from the command line	
Sep 17	7	– Introduction to pipelines – Shell scripts	
Sep 21	8	– Working on remote servers – High-performance computing	
Sep 24	9	– Introduction to R for data analysis	
Sep 28	10	– Introduction to Python for data analysis	
Oct 1	11	– *-Omics* research paper presentations	3-min talks + 2-min Q&A
Oct 5	12	– *-Omic*s research paper presentations	3-min talks + 2-min Q&A
Oct 8	13	– ***EXAM I***	
Oct 12*	14	– DNA sequencing methods	*Class meets on Tuesday, Oct 13 (Monday class schedule)
Oct 15	15	– Getting data from public repositories	
Oct 19	16	– Processing Illumina reads: QC, trimming, and filtering	
Oct 22	17	– Sequence alignment	
Oct 26	18	– Intro to BLAST and BLAST+	
Oct 29	19	– Genome and transcriptome assembly	
Nov 2	20	– Structural annotation of genomes and transcriptomes	
Nov 5	21	– Functional annotation of genomes and transcriptomes	
Nov 9	22	– Genome mapping and variant calling	
Nov 12	23	***EXAM II***	
Nov 16	24	– Phylogenetic inference from NGS data I	
Nov 19	25	– Phylogenetic inference from NGS data II	
Nov 23	26	– Machine learning in bioinformatics	
Nov 26	27		No class (Thanksgiving)
Nov 30	28	– Large language models in bioinformatics	
Dec 3	29	– Bioinformatics application paper presentations	10-min talks + 2-min Q&A
Dec 7	30	– Project repository presentations	10-min talks + 2-min Q&A
Dec 14	31	***FINAL EXAM***	Block Y (MR), Dec 14, 17:30 to 19:30

A Collection of Links for Useful Tools, Tutorials, Repositories, and Databases

Unix

- Unix basic commands: https://sandbox.bio/tutorials/terminal-basics/
- Unix shell: https://swcarpentry.github.io/shell-novice/01-intro.html
- Data exploration with awk: https://sandbox.bio/tutorials/awk-intro
- Loops: https://swcarpentry.github.io/shell-novice/05-loop.html

R

- Introduction to R: https://swirlstats.com
- Programming with R: http://swcarpentry.github.io/r-novice-inflammation/
- Data visualization in R using ggplot2: http://varianceexplained.org/RData/lessons/lesson2/

Python

- Introduction to Python: https://python.land/python-tutorial
- Programming with Python: https://swcarpentry.github.io/python-novice-inflammation/
- Data visualization in Python: https://python-graph-gallery.com

Genomic data processing and analysis

- Quality Control
  - FastQC: https://www.bioinformatics.babraham.ac.uk/projects/fastqc/
- Data Preprocessing
  - AdapterRemoval: https://adapterremoval.readthedocs.io/en/stable/
  - Trimmomatic: http://www.usadellab.org/cms/?page=trimmomatic
  - Cutadapt: https://cutadapt.readthedocs.io/en/stable/
  - SPAdes: https://cab.spbu.ru/software/spades/
  - BFC: https://github.com/lh3/bfc
- Genome Assembly
  - SPAdes: https://cab.spbu.ru/software/spades/
  - Canu: https://github.com/marbl/canu
  - BWA: http://bio-bwa.sourceforge.net/
  - Bowtie2: http://bowtie-bio.sourceforge.net/bowtie2/index.shtml
- Annotation
  - AUGUSTUS: http://bioinf.uni-greifswald.de/augustus/
  - GeneMark: http://exon.gatech.edu/GeneMark/
  - BLAST: https://blast.ncbi.nlm.nih.gov/Blast.cgi
  - InterProScan: https://www.ebi.ac.uk/interpro/search/sequence/
  - Pfam: https://pfam.xfam.org/
- Variant Calling
  - BWA: http://bio-bwa.sourceforge.net/
  - Bowtie2: http://bowtie-bio.sourceforge.net/bowtie2/index.shtml
  - GATK: https://gatk.broadinstitute.org/hc/en-us
  - FreeBayes: https://github.com/freebayes/freebayes
  - SAMtools: http://www.htslib.org/
- Comparative Genomics
  - MUSCLE: https://www.drive5.com/muscle/
  - MAFFT: https://mafft.cbrc.jp/alignment/software/
  - RAxML: https://cme.h-its.org/exelixis/web/software/raxml/
  - IQ-TREE: http://www.iqtree.org
  - PAML: http://abacus.gene.ucl.ac.uk/software/paml.html
- Functional Genomics
  - HISAT2: https://daehwankimlab.github.io/hisat2/
  - StringTie: http://ccb.jhu.edu/software/stringtie/

    - DESeq2: https://bioconductor.org/packages/release/bioc/html/DESeq2.html
    - Bismark: https://www.bioinformatics.babraham.ac.uk/projects/bismark/
    - MACS: https://github.com/macs3-project/MACS
- Data Visualization
    - UCSC Genome Browser: https://genome.ucsc.edu/
    - IGV (Integrative Genomics Viewer): http://software.broadinstitute.org/software/igv/
    - ggplot2 (R): https://ggplot2.tidyverse.org/
    - Matplotlib (Python): https://matplotlib.org/
- Data Collection, Storage and Management
    - NCBI: https://www.ncbi.nlm.nih.gov/
    - EMBL-EBI: https://www.ebi.ac.uk/